

JAYACHANDRAN JESSE JOHN

Engineering Manager — Cloud Infrastructure • Observability • ML Systems

Fremont, CA • (408) 807-1696 • jessejohn2k@gmail.com • [LinkedIn](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Engineering leader with a proven record of building and scaling cloud-native, data-driven platforms — now focused on production-grade AI systems. Experienced in designing multi-agent LLM pipelines, autonomous reasoning systems, and scalable Databricks architectures that turn large-scale operational data into real-time decisions. Managed multi-disciplinary teams across backend, data, and ML to deliver resilient systems supporting 100+ distributed deployments, with a reputation for incident leadership, system reliability, and cross-functional execution.

CORE SKILLS

AI Systems: LLMs, Multi-Agent Systems, Prompt Orchestration, RAG, Vector Search

AI-Assisted Development: Claude, Cursor (code review, refactoring, rapid prototyping)

Data Platforms: Databricks, Delta Lake, Unity Catalog, PySpark, ETL Pipelines

Cloud & Infra: AWS (EKS, Lambda, S3), Kubernetes, Distributed Systems

Programming: Python, SQL

Observability: Prometheus, Grafana, Loki, Elasticsearch

PROFESSIONAL EXPERIENCE

Tarana Wireless, Milpitas, CA

Software Engineering Manager | 12/2022 - Present

- Built an AI-driven autonomous Root Cause Analysis (RCA) platform on Databricks (Delta Lake, Unity Catalog, PySpark, Auto Loader); designed a multi-agent LLM system (Claude via ai_query) with prompt orchestration, confidence-based reasoning, self-refinement loops, and RCA result caching; integrated vector search (BGE embeddings) and knowledge-base retrieval to enable automated root-cause discovery and natural-language interaction with manufacturing data. Project Links: [NotebookLM Demo](#) | [Autonomous RCA System \(Architecture PDF\)](#)
- Architected and deployed a cloud-native test and analytics platform on AWS (EKS, Lambda, S3), integrating MES and factory systems for real-time production insights.
- Drove automation across 100+ distributed systems, improving stability, throughput, and reliability.
- Adopted Claude and Cursor across the engineering team for code review and development; established team practices for AI-assisted workflows, accelerating feature delivery and improving code quality.
- Led and mentored engineers, establishing design reviews and system-level debugging practices.
- Directed cross-functional incident response across hardware, software, and ML systems.

Oracle, Redwood City, CA

Principal Member of Technical Staff | 01/2017 - 12/2022

- Owned end-to-end delivery of critical PCA components—architecture, implementation, documentation, and handover to support—improving release readiness and reducing post-GA support burden.
- Delivered a [Kubernetes observability stack](#) (Prometheus, Fluentd, Loki, Grafana) for PCA 3.0.1, enabling proactive incident response and deep operational visibility.
- Designed a [next-generation Ethernet network architecture](#) for PCA 2.4.1 to improve high availability and business continuity across multi-rack deployments.
- Implemented the Switch Network Health Check (PCA 2.4.2) with automated diagnostics, strengthening reliability and reducing troubleshooting time.
- Contributed to SDN infrastructure; integrated with Cisco Nexus, Juniper QFX, and Arista top-of-rack (ToR) switches; enabled containerized workloads with Docker/Kubernetes.

- Provided technical leadership and mentorship across releases; drove code quality, agile execution, and efficient delivery for complex features.

Sonus Networks, Fremont, CA

Principal Engineer | 04/2015 – 12/2016

- Shipped SDN integration between Skype for Business Manager and the Vello/NaasIQ Controller with fault tolerance and high availability for enterprise voice and video.
- Built a modular Python test-automation framework (structured logging with dictConfig; reusable libraries) that accelerated regression cycles and validation.
- Modeled complex switch and traffic topologies with Mininet to validate controller performance under realistic, high-load conditions.
- Validated OpenFlow behavior and implemented Splunk-based telemetry for the Farsight app, improving diagnostics and visibility.

IBM, San Jose, CA

Staff Software Engineer | 08/2012 – 04/2015

- Developed automation and API-level testing for IBM's SDN VE Controller; integrated the OpenStack ML2 plugin for seamless network orchestration.
- Automated Layer 2, Layer 3, and multicast conformance with IxANVL, uncovering critical defects and improving standards compliance before GA.
- Built virtualized test environments (OpenStack, VMware, KVM) for repeatable control- and data-plane validation across diverse scenarios.
- Authored CLI and REST test suites; executed performance and stress tests with Ixia and Linux tooling to assess scalability and reliability.

Cisco Systems Inc., San Jose, CA

Hardware Engineer | 07/2007 – 12/2011

- Verified data-center switching platforms (MDS 9000; Nexus 2K/3K/5K; Catalyst 2K–6K), contributing to a portfolio exceeding \$1B in revenue.
- Authored feature, system, and regression plans with 1G/10G/40G SFP and copper interface verification; used Spirent, IXIA, SmartBits, and Agilent SAN testers for benchmarking.
- Collaborated across hardware and software teams on NX-OS/IOS bring-up; received the Cisco Achievement Performance Award for on-schedule delivery of an emerging-technology project.

EDUCATION & CERTIFICATIONS

- Data Analytics and Virtualization — UC Berkeley Extension (Professional Certification)
- M.S., Telecommunication Engineering Technology — Rochester Institute of Technology (Redcom Laboratories Scholarship; International Student Scholarship)
- B.S., Electronics & Telecommunication Engineering — Atharva College of Engineering, Mumbai University
- Certifications: Python • Red Hat Linux I & II • Mirantis OpenStack Admin (MCA200)

VOLUNTEERING

- Junior Team Tennis Spring Captain — USTA Northern California (Mar 2025 – May 2025); led team through [Spring Sectionals](#), organizing practice and mentoring junior players.
- Videographer — Resonate Movement (Jun 2022 – Jul 2025); produced video content for a local church.